

on copies of work indicates the copyright ownership. For instance, Monta Vista sued Lineo for distributing software written by the former without the copyright notice.

- Failure to provide licence notice. A licence notice contains the open source licence text included in the product or stack, and is usually provided with product documentation and/or within the product or application user interface. For example, the MIT licence notice reads: “The above copyright notice and this permission notice shall be included in all copies or substantial portions of the software.”
- Failure to provide attribution notice. An attribution notice has to be provided as a text file together with the open source component that provides the acknowledgement as supplied by the contributors of open source components.
- Failure to provide modification notice. A modification notice calls out modifications to the source code in a change log file. For instance, GPLv2 states: “You must cause the modified files to carry prominent notices stating that you changed the files and the date of any changes.”
- Failure to provide the source code. Certain licences require the source code to be provided. In the *Welte vs Fortinet UK Ltd. (2005)* suit, a preliminary injunction was sought against Fortinet prohibiting distribution of its products in the absence of a GPL code. Fortinet eventually agreed to make the source code in its products available under GPL.
- Failure to provide a written offer to end users with open source code in case of GPL/LGPL.
- Failure to provide scripts needed to compile the source code in case of GPL/LGPL.

Code scan

The code should be scanned and approved each time a developer modifies a previously approved component in a different context. Licences can change between different versions of the same inherited software component.

For example, React was initially licensed under Apache 2.0. In October 2014, React 0.12.0 replaced this with the 3-clause BSD licence and added a separate patent text file. Finally, on September 26, 2017, React 16.0.0 was released under the MIT licence.

Global trends in open source compliance enforcement

Community enforcement: The first set of open source enforcement cases were filed by community enforcers. From 2007 to 2009, the Software Freedom Law Center filed suits on behalf of Busy Box Principal Developers against various entities including Samsung, Verizon, BestBuy and Westinghouse for violations of the GNU GPLv2. The sole objective of these litigations was to ensure compliance.

Similar instances of community enforcement were also seen in Europe. Notable amongst the cases is the *gpl-violations.org* suit against D-Link in the District Court of Frankfurt in 2006.

Company/competitor enforcement: CoKinetic Systems Corporation sued its competitor Panasonic Avionics Corporation claiming US\$ 100 million for violation of GPLv2. GPLv2 requires reciprocal disclosure of the source code. Panasonic had allegedly refused to distribute the source code, and in turn, was accused of blocking its competitors (CoKinetic and others) from developing software for in-flight entertainment, along with other malpractices to monopolise the business. CoKinetic alleged that Panasonic’s refusal to distribute the source code was a blatant violation of the rights of numerous developers who have contributed to Linux. The case was eventually settled out of court.

Copyright troll enforcement:

Patrick McHardy, an erstwhile contributor to a Linux project, has had the reputation of profiting from copyright claims. Copyright trolls are on the lookout for companies which, often unknowingly, violate the terms of GPL. Such violations are bound to happen when companies embed third-party software into their own services and products without realising that they need to buy licences before using the software.

Dual-licence product enforcement:

Businesses often use dual-licence software. Such licensing structures allow them to use/modify/distribute the software with some restrictions such as the requirement to comply with an open source licence, or without those restrictions upon payment of a fee to obtain a commercial licence. Sometimes, the open source licence for dual-licence software prohibits its commercial use and distribution without a commercial licence. Businesses may mistakenly believe that since the software is available under the open source licence for personal use or hobbyists, it can also be used under the open source licence for commercial purposes. This leads to numerous lawsuits, most of which never end well for the infringer. Artifex, the developer of Ghostscript, an open source PDF interpreter, filed a suit against Hancom, a Korean developer of office apps. Hancom had used Ghostscript for developing and distributing an application without purchasing a commercial licence. No surprises here – Hancom lost the lawsuit.

User enforcement: End users of a product, which has an underlying open source code protected under GPL, may be able to seek open source compliance enforcement. For instance, AFPA sued Edu4 with the help of an open source advocacy group, FSF France, to enforce GPL after Edu4 refused to make its software source code available. Generally, only copyright owners